Action-Conditioned Grounding: Decoupling What to Do from Where to Act in GUI Agents

Anonymous Author(s)
Affiliation
Address
email

Abstract

[TODO: 150–200 words. Structure: (1) GUI agents must ground language intents to screen coordinates; errors are localization failures, a core grounded-perception problem. (2) We test whether factoring the policy into WHAT (action-type classification) and WHERE (action-type-conditioned grounding) improves grounding in a 2B-parameter VLM (Qwen2-VL-2B + LoRA). (3) Findings: action-type supervision helps when type is spatially predictive; the literal learned action-embedding mechanism is NOT the winning mechanism (auxiliary loss captures the benefit more simply); and a cautionary mechanistic result: naive inputs_embeds injection silently bypasses M-RoPE and produces false negatives. (4) Controlled multi-seed, low-data study relevant to practitioners deploying small VLMs.]

1 Introduction

[TODO: Frame GUI-action grounding as evidence localization for real-world deployment. The factorization hypothesis: separate "what action" from "where". Preview the supported/refuted/mechanism findings honestly — the honest-negative-result angle is the paper's personality. End with an explicit contribution list (3–4 bullets).]

2 Related Work

[TODO: Four short paragraphs: (1) GUI agents and screen grounding (Mind2Web, AITW, SeeClick, CogAgent, ScreenAI, OS-Atlas, UGround, Aguvis...); (2) action/goal-conditioned policies and factorized decision-making; (3) grounding + faithfulness in VLMs (workshop framing); (4) parameter-efficient adaptation of small VLMs. references.bib currently has only 15 entries — needs expansion to 30–40.]

3 Method: Factoring the GUI Policy

3.1 Problem Setup and Action Taxonomy

[TODO: UnifiedStep over Mind2Web + AITW; canonical 8-class action space; coordinates on a 0–999 grid emitted as text tokens.]

3.2 Stage 1: What — Action-Type Classification

[TODO: Frozen Qwen2-VL-2B pooled features + 3-layer MLP; text vs vision vs vision+text feature modes.]

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[TODO: Architecture figure: two-stage pipeline + the five Stage-2 conditioning variants.]

Figure 1: [TODO: caption]

3.3 Stage 2: Where — Action-Conditioned Grounding

[TODO: The five matched-compute conditioning variants: A (flat, no conditioning), B (auxiliary action-type loss), C (hard routing), D-hook (additive embedding-hook conditioning), D-token (M-RoPE-correct prepended learned action token). Figure 1: architecture diagram (NEEDS DRAWING).]

4 Experimental Setup

[TODO: Datasets and splits; data mixes (all_with_coords vs the taps_and_swipes control); metrics (hit@r, macro-F1); seeds 42/43/44; paired bootstrap over shared test examples; LoRA + training config; compute (single L4, total spend). Emphasize matched-compute comparisons.]

5 Results

5.1 Does Action-Type Supervision Improve Grounding?

[TODO: Main table: A/B/C/D-hook/D-token on all_with_coords, hit@0.10 mean $\pm$ std over 3 seeds + paired-bootstrap deltas vs A. Headline: B $\approx$ D-hook $>$ D-token $\gtrsim$ C $>$ A.]

5.2 Control: When Action Type Is Not Spatially Informative

[TODO: taps_and_swipes mix: conditioning neutral-to-harmful. This is the faithfulness/honesty angle — conditioning helps only when type carries spatial evidence.]

5.3 End-to-End: Predicted Types vs Oracle

[TODO: Stage-1 predicted types feeding Stage 2 beat flat A; $\sim$0.02 oracle gap.]

5.4 Low-Data Regime

[TODO: n_train $\in \{300, 500, 800\} \times$ seeds 42/43/44 for A/B/D-hook; D-hook most stable. Figure: scaling curves (results/phase4 figures).]

6 Mechanism Analysis

6.1 A Cautionary Tale: Embedding Injection Silently Breaks M-RoPE

[TODO: The false 8σ negative from inputs_embeds slot replacement; why Qwen2-VL's multimodal position ids depend on input_ids; D-hook/D-token fix. This subsection is a deployment-relevant failure-analysis contribution in its own right.]

6.2 Is the Learned Action Embedding Causally Used?

[TODO: D-token gold/wrong/zero intervention: gold $-$ wrong $=$ $+0.192$ hit@0.10, gold $-$ zero $=$ $+0.093$. "Used but not best", not "ignored".]

7 Discussion and Limitations

[TODO: Small model, small n, two datasets, coordinate-text output format; what generalizes and what might not. Honest scoping.]

8 Conclusion

[TODO: 3–4 sentences.]

References

A Reproduction Details

[TODO: Hyperparameters, prompts, data filtering, per-class metrics.]

B Additional Results

[TODO: Full per-seed tables; attention visualizations (results/phase4/attn_viz); Dfrozen post-mortem.]